

Micro Energy Harvesting 2026

Exercise 1a: Energy, power, energy forms and conversion

We want to start slowly, so this first exercise will introduce you to forms of energy, expected amounts of energy, power and power consumption.

Task 1: Energy

What was energy again?

Oh yes, there is something like potential energy, kinetic energy (& rotational energy), chemical energy, thermal energy, electrical energy, relativistic energy, quantum mechanical energy, etc.

- a) Identify the formulas for the above-mentioned forms of energy (if possible) and write them down together with an example and a typical energy value.

$$1 \text{ Joule} = 1 \text{ W} \cdot 1 \text{ s} = 1 \text{ N} \cdot 1 \text{ m} = 1 \text{ kg m}^2/\text{s}^2$$

Formula	Description	Example	Typ. Value
$E_{\text{pot}} = m \cdot g \cdot h$	mass * gravitational acceleration * height	30kg, 1,5m	450 J
$E_{\text{kin}} = 1/2 m v^2$	$\frac{1}{2} \cdot \text{mass} \cdot (\text{velocity})^2$	Car (1000kg, 100km/h)	385kJ
$E_{\text{kin,rot}} = 1/2 I \omega^2$	$\frac{1}{2} \cdot m \cdot \text{inertia} \cdot (\text{ang. velocity})^2$		
$E_{\text{chem}} = H = U + pV$ bzw. $\Delta H_R = \Delta H_{f, \text{Prod}} - \Delta H_{f, \text{Ed}}$	„enthalpy“ ↔ enthalpy of combustion (heating value)	1l Oil (refined)	40MJ
$E_{\text{therm}} = C_{\text{therm}} \cdot m \cdot T$	heat cap. * mass * temp.	1l H ₂ O 20°C -> 100 °C	390 kJ
$E_{\text{el}} = U \cdot I \cdot t$	voltage * current * time	1000W, 1h	3600 kJ, 36 MJ, 1kWh
$E_{\text{rel}} = m \cdot c^2$	mass * speed of light ²	Electro-positron pair rest mass: $2 \cdot 10^{-30} \text{ kg}$	2 x 511 keV (2 gamma quants) 0,16 pJ
$E_{\text{quant}} = h \cdot \nu$	Planck-const. * Frequency	$\lambda = 633 \text{ nm}$	$3,14 \cdot 10^{-19} \text{ J}$ (1,96 eV)

- b) How are energy and power related?

$$E = P \cdot t$$

- c) If you remember back to the first semester Physics. How much energy is expended when you lift a mass of 1 kg by one meter?

$$E = m \cdot g \cdot h = 1 \text{ kg} \cdot 10 \text{ m/s}^2 = 10 \text{ J (kg} \cdot \text{m}^2/\text{s}^2)$$

- d) And - let's assume you lose this energy again when you lower the mass again - how much power do you generate when you lift it every 10 seconds?

$$P = E/t = 10 \text{ J}/10 \text{ s} = 1 \text{ W}$$

e) Calculate the physical performance of a person with a body weight of 75 kg who jumps 10 cm into the air twice per second during boxing training (rope skipping). Also calculate the total energy consumption if you keep this up for 30 minutes.

$$P = E/t = 75 \text{ kg} * 10 \text{ m/s}^2 * 10 \text{ cm} * 2 / 1 \text{ s} = \underline{150 \text{ W}}$$

$$E = P * t = 150 \text{ W} * 1800 \text{ s} = 270 \text{ kWs} = \underline{75 \text{ Wh}}$$

Task 2: Energy content of materials, thermal energy, grey energy

In the first lecture the manufacturing energy for a copper cable with PVC sheathing was calculated and compared to the energy demand of a typical sensor. Comparing is fun, so you can compare some other things energetically!

a) Look up the energy content of the following things from literature: (The unit of energy is Joule J)

Heating value:

- Small serving of chips (1256 kJ)
- 1 L Diesel (40 MJ)
- 1 L Ethanol (24 MJ)
- 1 kg Hard coal (30 MJ)
- 1 stere of air-dry (15 % residual moisture) beech firewood (1900 kWh/rm, 6840 MJ)
- Heating value: approx. 10 % lower, as moisture does not play a major role
www.energie-experten.org
- Gas (36-50 MJ/kg)
- Hydrogen - H₂ (141.8 MJ/kg)

b) Find the energy required to heat the following items from 20 °C (293 K) to the specified temperature.

Thermal energy:

- 1 drop (=0.05 ml) of water, heated to 30 °C (~2 J)
- 1 drop of alcohol, heated to 30 °C (~1 J)
- 1 drop of copper at melting point (~183 J)
- 1 drop of aluminum at melting point (~77,4 J)
- 1 drop of tungsten at melting point (~422 J)

c) Determine the energy required to produce the following substances (difficult for some).

Gray (Embodied) energy:

- 1 kg silicon, wafer grade (~50,4 MJ)
- 1 kg aluminium (~155 MJ inkl. 33 % recycling)
210 MJ without recycling, 24 MJ with 100% recycling
https://en.wikipedia.org/wiki/Embodied_energy
- 1 kg copper (~42 MJ inkl. 37 % recycling)
~55 MJ without recycling
https://en.wikipedia.org/wiki/Embodied_energy
- 1 kg gold (~30 GJ)
- 1 kg glass (e.g. drinking glass) (~25,2 MJ)
- 1 L beer (Pils) (0,56 kWh = 2 MJ)

- 1 car (~75 GJ conventional; ~125 GJ electrical)
average car in Australia 270 GJ
https://en.wikipedia.org/wiki/Embodied_energy
- 1 pound of coffee (~27 MJ)
<http://dx.doi.org/10.1016/j.jclepro.2014.02.006> - Journal of Cleaner Production 73 (2014) 310-321
- 1 € (per issued) (~3,6 MJ)

Task 3: Typical input power for some devices

Estimate the power consumption of the following technical devices by internet research (additional task: try to estimate the values in advance - note the difference!)

Typical input power:

- LED green, super bright @operating voltage ($P=U \cdot I = 3.2 \text{ V} \cdot 20 \text{ mA} = 64 \text{ mW}$)
- CREE LED, MK-R, neutral white ($P=U \cdot I = 12 \text{ V} \cdot 1250 \text{ mA} = 15 \text{ W}$)
- Bike light (no matter where mounted / illuminant) (0,6 W, LED)
- Vacuum cleaner (100-500 W Akku / 1200-2400 W Kabel)
- Smartphone (during transmission, sending and display) ((Samsung Galaxy S5) Standby: 10,7 mW; Senden: 15 mW (GSM) – 22 mW (LTE) und 18 mW (WLAN); Vollast: 2 W)
<https://www.heise.de/ct/ausgabe/2015-22-Was-verbraucht-wie-viel-beim-Smartphone-2826274.html>
- Cortex M0+ microprocessor at 20 MHz clock frequency (11,2 μW pro MHz -> 224 μW , im High-Power Mode)
<https://www.silabs.com/products/mcu/32-bit/arm-cortex-m0-32-bit-microcontroller>
- MSP 430 μC with FRAM (5529) in deep sleep mode (LPM4.5: 0,18 μA @ 3,0 V = 0,54 μW)
Active Mode (AM): 290 $\mu\text{A}/\text{MHz}$ at 8 MHz, 3.0 V = 6,96 mW
<http://www.ti.com/product/MSP430F5529>
- Arduino Uno at 5 V under full load (46,5 mA @ 5 V = 232,5 mW)
34,4 mA in sleep mode (172 mW)
<https://www.gadgetmakersblog.com/arduino-power-consumption/>
- LED weiß, superhell @ Betriebsspannung ($P=U \cdot I = 3.6 \text{ V} \cdot 20 \text{ mA} = 72 \text{ mW}$)
- Ihr Fernseher (80 W)
- Ihr Computer (Notebook: 15 W – Gaming PC: 1000 W)
- Ihr Auto (40-100kW)